

Lesson 62 Practice: Answers and Rationales

JKSSB Computer Awareness

October 10, 2026

Provenance: All 25 items are original exam-pattern questions (the *[Original]* items plus the extension items tagged *[JKSSB FAA-pattern]*, *[JKPSI-pattern]*, *[SSC CGL-pattern]* and *[JKSSB-pattern]*, including MSQs); Concepts are drawn from PYQ-18/PYQ-19/PYQ-20; none reproduces an official paper item.

- Q1. B.** TCP stands for Transmission Control Protocol. A, C, and D are not the correct expansion.
- Q2. A.** TCP gives ordered and reliable delivery with no timing promise (PYQ-18). B invents a timing guarantee and drops ordering; C describes UDP; D describes ICMP.
- Q3. C.** UDP is connectionless and fast with neither ordering nor reliability. A moves a TCP property onto UDP; B reverses TCP into connectionless; D describes ICMP, not TCP.
- Q4. A.** ICMP stands for Internet Control Message Protocol. B, C, and D are not the correct full forms.
- Q5. C.** ICMP never transfers application data such as files or web pages; it carries control and diagnostic messages only (TRAP-19). A, B, and D are all true of ICMP.
- Q6. B.** BGP stands for Border Gateway Protocol. A, C, and D are not the correct expansion.
- Q7. A.** BGP is the inter-domain protocol routing between autonomous systems (PYQ-19, TRAP-20). B describes an interior protocol; C describes a mail protocol; D denies that BGP carries routing information.
- Q8. A.** BGP is inter-domain while OSPF and RIP are intra-domain. B swaps the scopes; C mislabels all three; D claims routing protocols transfer application data.
- Q9. C.** IPv6 addresses are 128-bit long (PYQ-18, TRAP-18). A is the IPv4 size; B is an intermediate distractor; D doubles the correct size.
- Q10. A.** IPv6 has a fixed 40-byte header and no broadcast address (multicast replaces it). B gives the IPv4 header size and a broadcast claim; C uses the 256-bit distractor with a broadcast; D mislabels IPv6 as a routing protocol.
- Q11. A.** The 128-bit space is so large that address-sharing via NAT is unnecessary (PYQ-19). B, C, and D name mechanisms unrelated to address supply.
- Q12. B.** HTTP/3 runs over QUIC over UDP, unlike HTTP/1.1 and HTTP/2 over TCP (PYQ-19, TRAP-21). A states the old stack; C describes ICMP; D describes BGP.
- Q13. A.** DNS stands for Domain Name System. B, C, and D are not the correct expansion.

- Q14. A.** DNS maps names to IP addresses hierarchically. B describes BGP; C describes POP3; D describes encryption.
- Q15. A.** PIR operates the .org registry while ICANN coordinates names and numbers globally. B swaps the two roles; C and D assign routing, mail, port, and diagnostic roles neither body holds.
- Q16. A.** ARPANET (1969) is the packet-switched ancestor of the Internet. B describes circuit-switched telephony; C describes IaaS; D describes a mail protocol.
- Q17. A.** Packet switching uses independent addressed packets while circuit switching holds one dedicated path. B swaps the two; C and D describe properties neither model has as stated.
- Q18. A.** Both the assertion and the reason are true, and reading the destination IP from the routing table is exactly how the router forwards between networks, so R correctly explains A. B denies the explanatory link; C and D each falsely mark one part false. A switch (MAC, one network) or a hub (repeats all) fits neither part.
- Q19. A.** Statements I and III are correct: TCP gives order and reliability without timing, and IaaS splits responsibility with networking hardware at the provider (PYQ-18, PYQ-20, TRAP-22). II is false because ICMP never transfers application data (TRAP-19). B selects only the false statement; C includes it; D pairs a true statement with the false one.
- Q20. B.** HTTP/3 uses QUIC over UDP and IPv6 is 128-bit with no broadcast, so both halves match. A swaps BGP inside a LAN and gives ICMP a file-transfer role; C gives the IaaS provider the customer OS and data and misstates PIR's role; D swaps the TCP/UDP properties.
- Q21. A.** IaaS is Infrastructure as a Service. B, C, and D are not the recognised expansion.
- Q22. A.** Providers own the physical layer in IaaS. B, C, and D are non-cloud chores.
- Q23. A.** IPv6 uses 128 bits. B is IPv4's length; C and D are not IP sizes.
- Q24. A, B, C.** IaaS, PaaS, and SaaS are the three models. D is not a service model.
- Q25. A, B, C.** The IaaS, SaaS, and BGP facts are correct. D is false: IPv6 is 128-bit.
- Q26. C.** IPv6 addresses are 128 bits. IPv4 addresses are 32 bits.
- Q27. C.** The Network layer provides logical addressing and routing. Data Link handles local frames; Session manages dialogues; Transport provides end-to-end services.
- Q28. A.** HTTP/3 uses QUIC over UDP. TCP is used by earlier HTTP versions; ICMP and ARP have other network roles.
- Q29. A.** IaaS provides infrastructure resources; the customer manages the guest OS and applications. SaaS delivers an application; PaaS provides a managed platform.
- Q30. A, B, C.** The first three are correct. QUIC runs over UDP, not TCP.